

Chandana Anand Rangappa

CA, United States | (408) 426-6291 | canan003@ucr.edu | [linkedin.com/in/chandanaaar](https://www.linkedin.com/in/chandanaaar)
github.com/Chandana2003 | chandana2003.github.io

EDUCATION

University of California, Riverside

Riverside, CA

Master of Science, Computer Engineering

Sep. 2024 – Mar. 2026

- **Relevant Coursework:** Distributed Systems, Operating Systems, Compilers, Software Testing & Analysis, AI, ML, Big Data Analytics, Information Retrieval, Data Mining, Database Management Systems, Real-Time Embedded Systems

Dayananda Sagar College of Engineering

Bengaluru, India

Bachelor of Engineering, Computer Science & Engineering

Sep. 2020 – Jun. 2024

- **Relevant Coursework:** Data Structures & Algorithms, Software Engineering, OOP in Java, Cloud Computing, Web Development, Machine Learning, NLP, Cyber Security, IoT, Computer Architecture, Computer Networks

EXPERIENCE

Graduate Researcher

Sep. 2025 – Mar. 2026

University of California, Riverside

Riverside, CA

- Built a pipeline using **Joern** and **Groq** Llama 3.3 70B to restore structural properties lost in **Ghidra** decompiled C code.
- Used agents.md to pass Joern-extracted hints as **RAG** context to the **LLM**, enabling recovery without exposing the source.
- Designed a self-improving workflow where a 2nd LLM agent reviews each session & updates agents.md with better rules.
- Applied chain-of-thought prompting and **3 specialist modules**, improving average recovery from **68.1%** to **81.4%**.
- Developed a scoring system to evaluate LLM recovery quality against ground truth, with best cases reaching **97.6%** recovery.

Software Engineering Intern

Dec. 2023 – May 2024

Anand Diagnostic Laboratory

Bengaluru, India

- Automated a fully manual antibiotic testing workflow using **YOLOv5**, **OpenCV**, and **TensorFlow**, achieving **95%** accuracy.
- Built a full-stack web app (**React**, **Flask**, **REST APIs**) with concurrent multi-user support, cutting reporting time by **30%**.
- Containerized the ML pipeline with **Docker** on **AWS**, set up **GitHub Actions** CI/CD, enabling reliable weekly deployments.
- Wrote **pytest** unit and integration tests ensuring data accuracy and reliability across all pipeline stages and API endpoints.
- Implemented **OAuth** auth flows with password hashing and protected API routes over **SSH**-secured infrastructure.

PROJECTS

AI-Assisted Semantic Property Discovery

- Built an agentic **LLM** pipeline using **Claude** and **Groq** APIs with **MCP** integrations to extract semantic properties.
- Generated test cases from extracted properties using **Hypothesis**, catching **60%** of introduced bugs across codebases.
- Evaluated LLM output quality across **250 programs**, identifying hallucination and over-generalization as failure patterns.

Academic Web Search Engine

- Built a BFS web crawler in **Python** indexing **500MB** across **50K+** .edu pages with SHA-256 deduplication.
- Designed a hybrid **BM25** and **BERT** search engine using **FAISS** and **Elasticsearch** for scalable semantic retrieval.
- Deployed on **GCP** with **Docker**, **Flask** REST API with **Redis** caching cut response time from **800ms** to **100ms**.

U-ASK: Spatial-Textual Search Engine

- Built a location-aware engine using **PostgreSQL** and **Elasticsearch** to query geospatial datasets by keyword and proximity.
- Implemented POWER and Boolean Range ranking for efficient **top-k** retrieval with negative keyword filtering.
- Optimized multi-dimensional query performance using spatial and inverted text indexing, improving query throughput by **95%**.

TECHNICAL SKILLS

Programming & Tools: Python, Java, JavaScript, TypeScript, C, C++, SQL, HTML, CSS, Git, Linux

Frameworks & Libraries: React.js, Angular, Flask, Node.js, GraphQL, PyTorch, TensorFlow, NumPy, Pandas, pytest

Software Engineering: DevOps, CI/CD, GitHub Actions, Concurrency, OOP, Microservices, Agile

Databases & Cloud: PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch, AWS, GCP, Docker, Kubernetes

AI & ML: LLMs, Agentic AI, RAG, Claude API, Groq API, FAISS, Computer Vision, NLP

PUBLICATIONS & RECOGNITION

Publications: **Inhibition Zones: A Window into Antibiotic Resistance** – ML pipeline for antibiotic testing

IoT Solutions for Food Quality Monitoring – ESP32 and sensor-based real-time food safety monitoring system

Recognition: Won Best Innovative Project Award (Nokia Open Day, DSCE) for the Antibiotic Susceptibility Testing System
Subcommittee Chair, Student Services Fee Advisory Committee – UC Riverside